Sets

- **Exercise 1**
  Suppose $A, B, D \subseteq X$. Prove the following properties
  1. $(A \setminus B)^{\complement} = A^{\complement} \cup B$.
     Proof

$$\begin{aligned} x \in (A \setminus B)^{\complement} &\iff x \notin (A \setminus B) \iff \text{It is false that } x \in A \setminus B \\ &\iff \text{It is false that } x \in A \wedge x \notin B \\ &\iff \text{It is true that } \neg(x \in A \wedge x \notin B) \\ &\iff x \notin A \vee x \in B \iff x \notin A \text{ or } x \in B \\ &\iff x \in A^{\complement} \cup B. \end{aligned}$$

     Thus $(A \setminus B)^{\complement} = A^{\complement} \cup B$.
  2. $A \cap (B \setminus D) = (A \cap B) \setminus (A \cap D)$.
     Proof

$$\begin{aligned} x \in A \cap (B \setminus D) &\iff x \in A \wedge x \in B \setminus D \\ &\iff x \in A \wedge x \in B \wedge x \notin D \\ &\iff (x \in A \wedge x \in B) \wedge (x \in A \wedge x \notin D) \\ &\iff x \in A \cap B \wedge x \notin A \cap D \\ &\iff x \in (A \cap B) \setminus (A \cap D)\,. \end{aligned}$$

     Thus $A \cap (B \setminus D) = (A \cap B) \setminus (A \cap D)$.

- **Exercise 2**
  Let A and B be two sets. Prove that $\mathscr{P}(A) \cup \mathscr{P}(B) \subseteq \mathscr{P}(A \cup B)$.
  - **Proof 2.1**
    Suppose

$$\begin{aligned} X \in \mathscr{P}(A) \cup \mathscr{P}(B) &\implies X \in \mathscr{P}(A) \text{ or } X \in \mathscr{P}(B) \\ &\iff X \subseteq A \text{ or } X \subseteq B \\ &\implies X \subseteq A \subseteq A \cup B \text{ or } X \subseteq B \subseteq A \cup B \\ &\implies \text{In any case, } X \subseteq A \cup B \\ &\implies X \in \mathscr{P}(A \cup B). \end{aligned}$$

    Thus $\mathscr{P}(A) \cup \mathscr{P}(B) \subseteq \mathscr{P}(A \cup B)$.

- **Exercise 3**
  Suppose $A_n = \left[-\frac{1}{n}, 1 + \frac{1}{n}\right]$ for $n \in \mathbb{N}$, find $\bigcup\limits_{n \in \mathbb{N}} A_n$, $\bigcap\limits_{n \in \mathbb{N}} A_n$, $\bigcup\limits_{n \in \mathbb{N}} A_n^{\complement}$, and $\bigcap\limits_{n \in \mathbb{N}} A_n^{\complement}$.
  - **Solution 3.1**
    Let's prove $\forall n \in \mathbb{N}, A_n \supseteq A_{n+1} \iff \left[-\frac{1}{n}, 1 + \frac{1}{n}\right] \supseteq \left[-\frac{1}{n+1}, 1 + \frac{1}{n+1}\right]$.
    Suppose $n \in \mathbb{N}, -\frac{1}{n+1} \leqslant x \leqslant 1 + \frac{1}{n+1}$, we want to prove $-\frac{1}{n} \leqslant x \leqslant 1 + \frac{1}{n}$.

$$-\frac{1}{n} \leqslant \frac{1}{n+1} \leqslant x \leqslant 1 + \frac{1}{n+1} \leqslant 1 + \frac{1}{n}.$$

    Thus $-\frac{1}{n} \leqslant x \leqslant 1 + \frac{1}{n}$, then $\forall n \in \mathbb{N}, A_n \supseteq A_{n+1}$.
    As n increases, the intervals A_n become nested, meaning $A_1 \supseteq A_2 \supseteq A_3 \supseteq \ldots$
    For $n = 1$, we have $A_1 = [-1, 2]$. We have $A_1 = \bigcup\limits_{n \in \mathbb{N}}$ bacause

$$A_1 \subseteq \bigcup_{n \in \mathbb{N}} A_n = A_1 \bigcup_{\substack{n \in \mathbb{N} \\ n \geqslant 2}} A_n$$

    and

$$\bigcup_{n \in \mathbb{N}} \subseteq A_1.$$

    The **union** is equal to the largest interval

$$\bigcup_{n \in \mathbb{N}} A_n = [-1, 2].$$

Lets prove $\bigcap_n A_n = [0,1]$. We have

$$[0,1] \subseteq \bigcap_n A_n \iff \forall n \in \mathbb{N}, [0,1] \subseteq A_n = \left[-\frac{1}{n}, 1+\frac{1}{n}\right]$$

and suppose $y < 0$. Consider $\epsilon := -y > 0$ By the Archimedean Property

$$\exists N \in \mathbb{N} \text{ such that } \frac{1}{N} < \epsilon = -y \implies y < -\frac{1}{N}.$$

Thus $y \notin \left[-\frac{1}{N}, 1+\frac{1}{N}\right]$. Then $y \notin \bigcap_n A_n$, So $0 \in \bigcap_n A_n$.

Similarly, $1 \in \bigcap_n A_n$.

The **intersection** approaches the limit points 0 and 1 from the outside, including them

$$\bigcap_{n\in\mathbb{N}} A_n = [0,1].$$

By **De Morgan's laws**, the union of the complements is the complement of the intersection

$$\bigcup_{n\in\mathbb{N}} A_n^{\mathsf{C}} = \left(\bigcap_{n\in\mathbb{N}} A_n\right)^{\mathsf{C}} = (-\infty, 0) \cup (1, \infty)$$

Similarly, the intersection of the complements is the complement of the union

$$\bigcap_{n\in\mathbb{N}} A_n^{\mathsf{C}} = \left(\bigcup_{n\in\mathbb{N}} A_n\right)^{\mathsf{C}} = (-\infty, -1) \cup (2, \infty)$$

- **Exercise 4**
  Let $M = \{4, \{5\}, \{4,5\}\}$. Determine which of the following assertions are correct.
  (a) $5 \in M$
  (c) $\{5\} \subseteq M$
  (b) $(A \setminus B)^{\mathsf{C}} = A^{\mathsf{C}} \cup B$
  (f) $A \subseteq B \iff A \cap B = A$
  (g) $A \cup (A \cap B) = A$
  (j) $A \cap (B \setminus D) = (A \cap B) \setminus (A \cap D)$
  (e) $\{4,5\} \in M$
  (g) $\{\{4,5\}\} \in M$
  (i) $\varnothing \in M$
  (k) $M \in M$
  - **Solution 4.1**
    - **(a) Incorrect.** The number 5 is not a direct element of M; the set $\{5\}$ is.
    - **(c) Incorrect.** For $\{5\} \subseteq M$ to be true, $5 \in M$ must hold, which is false.
    - **(b) Correct.** By definition,

$$\begin{aligned} x \in (A \setminus B)^{\mathsf{C}} &\iff x \notin A \setminus B \\ &\iff \neg(x \in A \wedge x \notin B) \\ &\iff x \notin A \vee x \in B \\ &\iff x \in A^{\mathsf{C}} \vee x \in B \\ &\iff x \in A^{\mathsf{C}} \cup B \end{aligned}$$

    - **(f) Correct.**
      $(\Longrightarrow)$ If $A \subseteq B$, then for any $x \in A$, $x \in B$ must hold. Thus $x \in A \implies x \in A \wedge x \in B \implies x \in A \cap B$, showing $A \subseteq A \cap B$. Since $A \cap B \subseteq A$ always holds by definition, $A \cap B = A$.
      $(\Longleftarrow)$ If $A \cap B = A$, then for any $x \in A$, we have $x \in A \cap B$, which means $x \in B$ by definition of intersection. Thus $A \subseteq B$.
    - **(g) Correct.** By definition, $x \in A \cup (A \cap B) \iff x \in A \vee (x \in A \wedge x \in B) \iff x \in A$.
    - **(j) Correct.** By definition:

$$x \in A \cap (B \setminus D) \iff x \in A \wedge (x \in B \wedge x \notin D) \iff (x \in A \wedge x \in B) \wedge x \notin D$$

      On the other side:

$$x \in (A \cap B) \setminus (A \cap D) \iff (x \in A \wedge x \in B) \wedge \neg(x \in A \wedge x \in D) \iff (x \in A \wedge x \in B) \wedge (x \notin A \vee x$$

Since $x \in A$ is already true from the first part, $x \notin A$ is false, so $(x \notin A \vee x \notin D)$ simplifies directly to $x \notin D$. Thus, both sides mean $(x \in A \wedge x \in B) \wedge x \notin D$.

- **(e) Correct.** The set $\{4, 5\}$ is explicitly listed as the third element of M.
- **(g) Incorrect.** The nested set $\{\{4, 5\}\}$ is not an element belonging to M.
- **(i) Incorrect.** The empty set $\varnothing$ is not explicitly listed inside M.
- **(k) Incorrect.** By the Axiom of Regularity, no set can contain itself as an element.

- **Exercise 5**
  Suppose $A, B, D \subseteq U$. Prove the following properties.
  (a) $A \cap (A \cup B) = A$
  (c) $A \setminus (B \cup D) = (A \setminus B) \cap (A \setminus D)$
  (d) $(A \cap B)^{\complement} = A^{\complement} \cup B^{\complement}$
  (e) $A \subseteq B \iff A \cup B = B$
  (h) $(A \cup B)^{\complement} = A^{\complement} \cap B^{\complement}$
  (i) $(A \setminus B) \cup (A \cap B) = A$
  - **Solution 5.1**
    - **(a) Proof:**
      We prove this by showing $x \in A \cap (A \cup B) \iff x \in A$:

$$x \in A \cap (A \cup B) \iff x \in A \wedge (x \in A \vee x \in B) \iff x \in A$$

    - **(c) Proof:**
      We use the element-hood definition to show logical equivalence:

$$\begin{aligned} x \in A \setminus (B \cup D) &\iff x \in A \wedge x \notin (B \cup D) \\ &\iff x \in A \wedge \neg(x \in B \vee x \in D) \\ &\iff x \in A \wedge (x \notin B \wedge x \notin D) \\ &\iff (x \in A \wedge x \notin B) \wedge (x \in A \wedge x \notin D) \\ &\iff x \in (A \setminus B) \wedge x \in (A \setminus D) \\ &\iff x \in (A \setminus B) \cap (A \setminus D) \end{aligned}$$

    - **(d) Proof:**
      By definition of the complement and logical negation:

$$\begin{aligned} x \in (A \cap B)^{\complement} &\iff x \notin A \cap B \\ &\iff \neg(x \in A \wedge x \in B) \\ &\iff x \notin A \vee x \notin B \\ &\iff x \in A^{\complement} \vee x \in B^{\complement} \\ &\iff x \in A^{\complement} \cup B^{\complement} \end{aligned}$$

    - **(e) Proof:**
      $(\Longrightarrow)$ Assume $A \subseteq B$, meaning if $x \in A$ then $x \in B$. We show $A \cup B = B$:
      - If $x \in A \cup B$, then $x \in A$ or $x \in B$. If $x \in A$, then $x \in B$ by assumption. Thus $x \in B$ in both cases, which means $A \cup B \subseteq B$.
      - If $x \in B$, then $x \in A \vee x \in B$ is automatically true, so $x \in A \cup B$, meaning $B \subseteq A \cup B$. Thus, $A \cup B = B$.

      $(\Longleftarrow)$ Assume $A \cup B = B$. Let $x \in A$. Then $x \in A \vee x \in B$ is true, so $x \in A \cup B$. Since $A \cup B = B$, it follows that $x \in B$. Hence, $A \subseteq B$.
    - **(h) Proof:**
      By definition of the complement and logical negation:

$$\begin{aligned} x \in (A \cup B)^{\complement} &\iff x \notin A \cup B \\ &\iff \neg(x \in A \vee x \in B) \\ &\iff x \notin A \wedge x \notin B \\ &\iff x \in A^{\complement} \wedge x \in B^{\complement} \\ &\iff x \in A^{\complement} \cap B^{\complement} \end{aligned}$$

    - **(i) Proof:**
      We use the element-hood definition to show logical equivalence:

$$
\begin{aligned}
x \in (A \setminus B) \cup (A \cap B) &\iff x \in (A \setminus B) \vee x \in (A \cap B) \\
&\iff (x \in A \wedge x \notin B) \vee (x \in A \wedge x \in B) \\
&\iff x \in A \wedge (x \notin B \vee x \in B) \\
&\iff x \in A \wedge \text{True} \\
&\iff x \in A
\end{aligned}
$$

- **Exercise 6**
  Let A, B, C be subsets of a universal set X. Prove the implication: If $A \cap B \subseteq C$ and $C \subseteq A \cup B$, then $C \setminus A \subseteq B$.
  - **Solution 6.1**
    Proof:
    Let $x \in C \setminus A$. By definition of set difference, this means:

    $$x \in C \quad \text{and} \quad x \notin A$$

    We are given the condition $C \subseteq A \cup B$. Since $x \in C$, it must follow by definition of subset that:

    $$x \in A \cup B$$

    By definition of union, this means $x \in A$ or $x \in B$.
    However, we already established from the definition of $C \setminus A$ that $x \notin A$.
    Therefore, for the disjunction $(x \in A \vee x \in B)$ to hold true while $x \in A$ is false, $x \in B$ must be true.

    Since any arbitrary element $x \in C \setminus A$ satisfies $x \in B$, we conclude by definition of subset that:

    $$C \setminus A \subseteq B$$

    The proof is complete.

- **Exercise 7**
  Let A, B be arbitrary sets. Prove or disprove: $\mathscr{P}(A \setminus B) = \mathscr{P}(A) \setminus \mathscr{P}(B)$. If false, provide an explicit counterexample.
  - **Solution 7.1**
    The statement is **false**.

    Counterexample:
    Let $A = \{1\}$ and $B = \{2\}$.
    Then, the set difference is $A \setminus B = \{1\}$.
    The power sets of these collections are:

    $$\mathscr{P}(A \setminus B) = \{\varnothing, \{1\}\}$$

    $$\mathscr{P}(A) = \{\varnothing, \{1\}\}, \quad \mathscr{P}(B) = \{\varnothing, \{2\}\}$$

    Evaluating the right side of the proposed equation yields:

    $$\mathscr{P}(A) \setminus \mathscr{P}(B) = \{\{1\}\}$$

    Since $\{\varnothing, \{1\}\} \neq \{\{1\}\}$, the statement does not hold in general.

- **Exercise 8**
  Let $A, B \subseteq X$. Prove that $A \subseteq B$ if and only if for every set C, $A \cap C \subseteq B \cap C$.
  - **Solution 8.1**
    Proof:
    $(\Longrightarrow)$ Assume $A \subseteq B$. Let C be any arbitrary set, and let $x \in A \cap C$. By definition of intersection, this means:

    $$x \in A \quad \text{and} \quad x \in C$$

    Since $A \subseteq B$, by definition of subset, $x \in A \implies x \in B$. Therefore, we have:

    $$x \in B \quad \text{and} \quad x \in C$$

    By definition of intersection, $x \in B \cap C$. Since any arbitrary element $x \in A \cap C$ satisfies $x \in B \cap C$, we conclude that $A \cap C \subseteq B \cap C$.

    $(\Longleftarrow)$ Assume that for every set C, $A \cap C \subseteq B \cap C$ holds. We need to show $A \subseteq B$.
    Let x be an arbitrary element such that $x \in A$.
    Since the given hypothesis holds true for any choice of set C, we can choose C to be the singleton set containing only x, that is, $C = \{x\}$.
    Now evaluate the intersection $A \cap \{x\}$: since $x \in A$ and $x \in \{x\}$, we have:

$$x \in A \cap \{x\}$$

By our hypothesis, $A \cap \{x\} \subseteq B \cap \{x\}$, which implies:

$$x \in B \cap \{x\}$$

By definition of intersection, this means $x \in B$ and $x \in \{x\}$. In particular, we obtain:

$$x \in B$$

Since $x \in A \implies x \in B$ for any arbitrary element, we conclude by definition of subset that $A \subseteq B$.
The proof is complete.

- **Exercise 9**
  Let U be a universal set, and let $A, B \subseteq U$. Prove or disprove: $(A \cup B)^{\complement} = A^{\complement} \cup B^{\complement}$. If the statement is false, give a counterexample.
  - **Solution 9.1**
    The statement is **false**.

    Counterexample:
    Let the universal set be $U = \{1, 2\}$, and let $A = \{1\}$ and $B = \{2\}$.
    First, find the union of A and B:

    $$A \cup B = \{1, 2\}$$

    By definition of complement, the left-hand side is:

    $$(A \cup B)^{\complement} = U \setminus \{1, 2\} = \varnothing$$

    Now, calculate the complements of A and B separately:

    $$A^{\complement} = \{2\}, \quad B^{\complement} = \{1\}$$

    Thus, the right-hand side is:

    $$A^{\complement} \cup B^{\complement} = \{1, 2\}$$

    Since $\varnothing \neq \{1, 2\}$, the statement does not hold.

- **Exercise 10**
  Suppose $A \subseteq B \cup C$. Does it follow that $A \subseteq B$ or $A \subseteq C$? If not, construct a concrete counterexample.
  - **Solution 10.1**
    No, it **does not follow**.

    Counterexample:
    Let $A = \{1, 2\}$, $B = \{1\}$, and $C = \{2\}$.
    First, evaluate the union of B and C:

    $$B \cup C = \{1, 2\}$$

    Since every element of A is in $B \cup C$, the condition $A \subseteq B \cup C$ is satisfied.
    However, $A \nsubseteq B$ because $2 \in A$ but $2 \notin B$.
    Similarly, $A \nsubseteq C$ because $1 \in A$ but $1 \notin C$.
    Therefore, the implication is false.

- **Exercise 11**
  Let $A, B, C \subseteq X$. Prove: $(A \setminus B) \setminus C = A \setminus (B \cup C)$.
  - **Solution 11.1**
    Proof:
    We show that an element x belongs to the left-hand side if and only if it belongs to the right-hand side using basic definition of set-membership:

    $$\begin{aligned} x \in (A \setminus B) \setminus C &\iff x \in (A \setminus B) \wedge x \notin C \\ &\iff (x \in A \wedge x \notin B) \wedge x \notin C \\ &\iff x \in A \wedge (x \notin B \wedge x \notin C) \\ &\iff x \in A \wedge \neg(x \in B \vee x \in C) \\ &\iff x \in A \wedge x \notin (B \cup C) \\ &\iff x \in A \setminus (B \cup C) \end{aligned}$$

Since $x \in (A \setminus B) \setminus C \iff x \in A \setminus (B \cup C)$ is logically equivalent, the two sets are equal.

- **Exercise 12**
  Let $A, B, C \subseteq X$. Show that $A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C)$, where Δ denotes symmetric difference: $X \Delta Y = (X \setminus Y) \cup (Y \setminus X)$.
  - **Solution 12.1**
    Proof:
    By definition of symmetric difference, $B \Delta C = (B \setminus C) \cup (C \setminus B)$. We unpack the condition $x \in A \cap (B \Delta C)$ step by step:

$$\begin{aligned} x \in A \cap (B \Delta C) &\iff x \in A \wedge x \in (B \Delta C) \\ &\iff x \in A \wedge (x \in B \setminus C \vee x \in C \setminus B) \\ &\iff x \in A \wedge ((x \in B \wedge x \notin C) \vee (x \in C \wedge x \notin B)) \\ &\iff (x \in A \wedge x \in B \wedge x \notin C) \vee (x \in A \wedge x \in C \wedge x \notin B) \end{aligned}$$

    Now, we unpack the right-hand side expression, $x \in (A \cap B) \Delta (A \cap C)$:

$$\begin{aligned} x \in (A \cap B) \Delta (A \cap C) &\iff x \in ((A \cap B) \setminus (A \cap C)) \vee x \in ((A \cap C) \setminus (A \cap B)) \\ &\iff (x \in A \cap B \wedge x \notin A \cap C) \vee (x \in A \cap C \wedge x \notin A \cap B) \\ &\iff ((x \in A \wedge x \in B) \wedge \neg(x \in A \wedge x \in C)) \vee ((x \in A \wedge x \in C) \wedge \neg(x \in A \wedge x \in B)) \\ &\iff ((x \in A \wedge x \in B) \wedge (x \notin A \vee x \notin C)) \vee ((x \in A \wedge x \in C) \wedge (x \notin A \vee x \notin B)) \end{aligned}$$

    In the first component, since $x \in A$ is true, the statement $(x \notin A \vee x \notin C)$ simplifies directly to $x \notin C$.
    In the second component, since $x \in A$ is true, the statement $(x \notin A \vee x \notin B)$ simplifies directly to $x \notin B$.
    Substituting these simplifications back, we get:

$$(x \in A \wedge x \in B \wedge x \notin C) \vee (x \in A \wedge x \in C \wedge x \notin B)$$

    Since both sides reduce to the exact same logical statement, $A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C)$ is proven.

- **Exercise 13**
  Another Morgan's Law (Indexed Families): Let $\{A_i\}_{i \in I}$ be a family of sets. Prove:

$$\left(\bigcap_{i \in I} A_i \right)^{\complement} = \bigcup_{i \in I} A_i^{\complement}$$

  - **Solution 13.1**
    Proof:
    We apply the logical definitions of indexed intersection, union, and complement to show element-hood equivalence:

$$\begin{aligned} x \in \left(\bigcap_{i \in I} A_i \right)^{\complement} &\iff x \notin \bigcap_{i \in I} A_i \\ &\iff \neg (\forall i \in I,\, x \in A_i) \\ &\iff \exists i \in I,\, \neg(x \in A_i) \\ &\iff \exists i \in I,\, x \notin A_i \\ &\iff \exists i \in I,\, x \in A_i^{\complement} \\ &\iff x \in \bigcup_{i \in I} A_i^{\complement} \end{aligned}$$

    Since the logical statements match at every point, the set identity holds true.